

ResQMesh

Disconnected-First Disaster Coordination Infrastructure

Team Name	BlindSpot
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Selected Track	Reinvent Digital Public Infrastructure For Billions

1. PROBLEM STATEMENT

- **Connectivity fails:** disasters can cut power and cellular/internet access, leaving a working phone unable to send an SOS.
- **Visibility fails:** available hospital beds, shelters, ambulances and community resources may exist but remain invisible to responders.
- **Maps go stale:** blocked roads and changing conditions can make static routing unsafe during a fast-moving emergency.
- **Trust breaks:** duplicate and contradictory reports make it difficult to determine what is current, corroborated and actionable.

Who faces it: survivors in rural, hilly and coastal areas; community volunteers; hospitals and shelters; rescue teams; and district/state disaster authorities coordinating with incomplete information.

Why it matters: the gap is not only a communication problem. It is an information and coordination gap between people who need help, resources that can help, and responders who need a trusted operational picture.

2. PROPOSED SOLUTION

ResQMesh is a disconnected-first Digital Public Infrastructure layer for disaster coordination. It combines **offline communication, distributed disaster knowledge, community resource matching, and information provenance** so coordination can continue even when conventional connectivity is unavailable.

Core flow: Citizen (offline) → peer relay (Wi-Fi/BLE) → rescue vehicle or nearby node regains connectivity → synchronization → knowledge graph + resource matching + route update.

Every SOS packet: unique ID · timestamp · priority · TTL · Ed25519 signature — enabling bounded relay, duplicate detection and authenticated messages.

- **Offline SOS Mesh:** store-and-forward, hop-by-hop relay with priority, TTL and duplicate detection.
- **Distributed Knowledge Graph:** each node maintains local relationships among people, needs, resources, roads and responders; event-based synchronization merges updates on reconnection.
- **Community Resource Mesh:** matches Need → Capability → Location → Availability → Assignment, making community assets discoverable.
- **Provenance & Trust Engine:** tracks source, timestamp, freshness and corroboration; conflicting observations are weighted rather than blindly accepted.

ResQMesh | System Architecture

ResQMesh

System Architecture

Disconnected-first disaster coordination infrastructure

Public infrastructure for communication, resource coordination and trusted situational awareness — even when the network is gone.

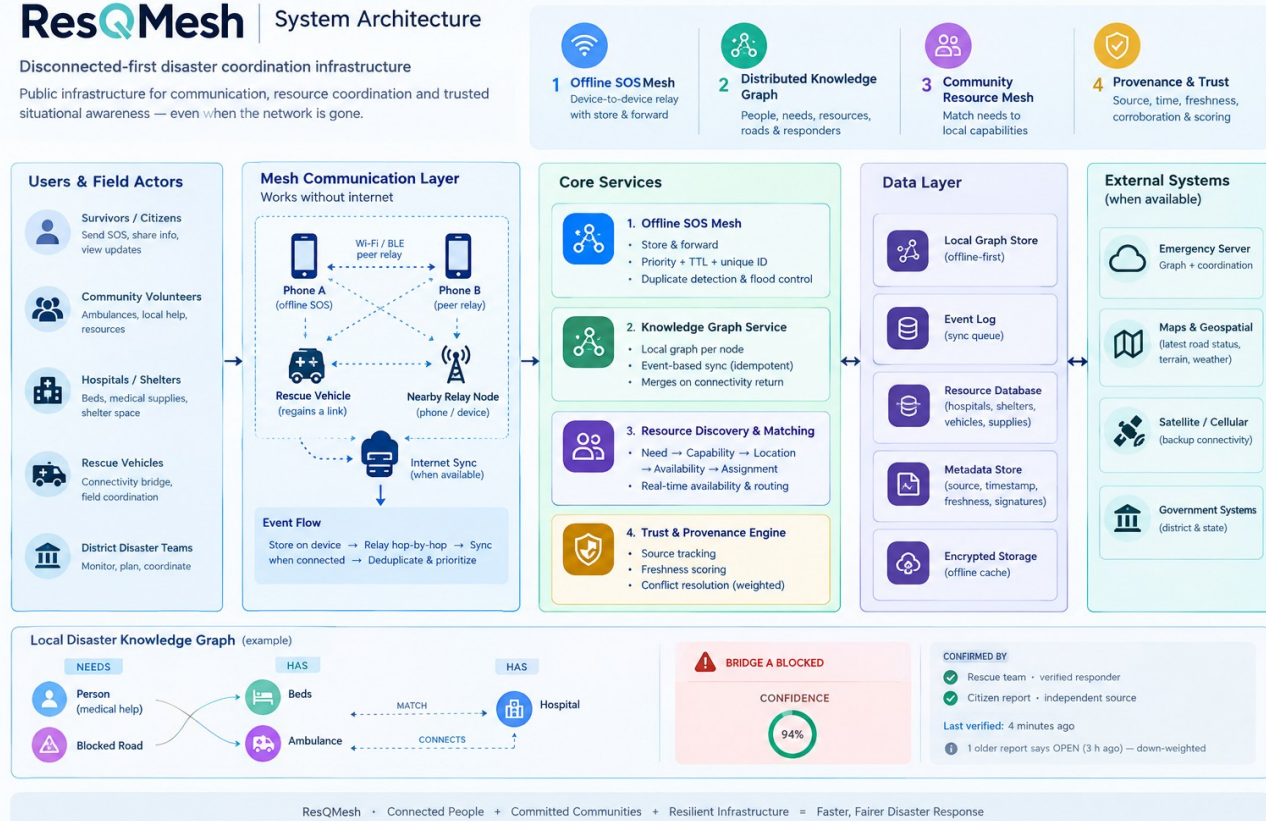


Figure 1. ResQMesh architecture: Users & Field Actors → Mesh Communication Layer → Core Services → Data Layer → External Systems.

Core Architecture

- **Mesh Communication Layer:** offline peer relay using store-and-forward; prototype uses WebSocket over local Wi-Fi/LAN, with BLE/Wi-Fi Direct as the native-app roadmap.
- **Core Services:** Offline SOS Mesh, Knowledge Graph Service, Resource Discovery & Matching, and Trust & Provenance Engine.
- **Data Layer:** local graph storage, event/sync queue, resource database, metadata/provenance storage and encrypted offline cache.
- **External Systems:** emergency servers, maps/geospatial services, satellite/cellular connectivity and government systems when connectivity is available.

3. TARGET USERS & IMPACT

- **Survivors / citizens:** one-tap SOS, information sharing and updates without requiring an active internet connection.
- **Community volunteers:** ambulances, generators, medicine stock, local help and other capabilities become discoverable resources.
- **Hospitals / shelters:** bed, capacity and supply status can be represented in the coordination layer.
- **Rescue vehicles / district teams:** regain connectivity bridges and coordinate needs, resources and routes from one operational picture.

Designed for the underserved: no account is required to raise an SOS; payloads are lightweight and the architecture targets low-end devices and low-connectivity environments. The same infrastructure can support floods, cyclones, landslides, earthquakes, wildfires and urban emergencies.

4. TECHNOLOGY STACK

Client	React PWA · Service Worker · IndexedDB · Leaflet · OpenStreetMap
Peer / Sync	WebSocket over Wi-Fi/LAN (prototype) · BLE / Wi-Fi Direct via native app (roadmap) · event-sourced log · UUID + TTL
Backend	Python · FastAPI · WebSocket · Pydantic
Data	PostgreSQL + PostGIS · pgRouting / OSMnx · Neo4j knowledge graph
Security & Interop	Ed25519 signatures · encrypted sensitive fields · OASIS CAP 1.2 · GeoJSON

5. INNOVATION & SCALABILITY

- **One infrastructure layer:** integrates disconnected-first communication, distributed knowledge, resource discovery and provenance instead of solving only one failure mode.
- **Many-to-many coordination:** citizens, responders, hospitals, shelters and community resources participate as connected nodes.
- **Trust-aware information:** reports carry provenance and freshness so the system can handle conflicting observations.
- **Scalable federation:** TTL- and priority-bounded relay limits traffic; graph partitioning by district/geohash and idempotent synchronization support expansion from neighbourhood → district → state → national federation.
- **Interoperability:** CAP-compatible alerts, GeoJSON and open mapping data allow integration with existing emergency infrastructure.

6. PROTOTYPE / REFERENCES

Wireframe/demo: <https://tinyurl.com/hackathon223>

Prototype plan: the 24-hour MVP demonstrates one complete scenario: offline SOS → relay across simulated nodes → reconnection → synchronization → graph-based resource matching → route update after a road-status change.

References: OpenStreetMap · OASIS Common Alerting Protocol (CAP) v1.2 · RFC 4838 Delay-Tolerant Networking Architecture · National Disaster Management Authority (India).

7. TEAM CONTRIBUTION

Supreet Vardhamane	Offline SOS Mesh
Saroj Ghosh	System Architecture & Integration
Satyakam Vats	Distributed Knowledge Graph
Nanditha L M	Resource Discovery & Matching
Jahanvi Pagadala	Client / UI (React PWA)
Yashika Arora	Provenance & Trust Engine

Core proposition: When centralized infrastructure fails, ResQMesh turns nearby devices, people and community resources into a resilient, trusted coordination network.